

Lab Objective:

- Learn MATLAB Application and do simple mathematic calculation problems.

Lab Preparation:

- Read the course slides and chapter 1 of the textbook.

Lab Procedures:

1. Open the MATLAB software.
2. In the command window try to enter a number like 190 and see what happens.
3. In the command window try to enter simple expressions like $8*7$ or $100+12$ and see what happens.
4. In the command window enter the following: `num1 = -1`.
5. In the command window enter the following: `num2 = 90`.
6. Now try to enter `num3 = num1 / num2` in the command window and see what happens.
7. Now try to enter `num4 = num1 * num2` in the command window and see what happens.
8. Use the **workspace window** to observe the results of `num1`, `num2`, `num3`, and `num4`.
9. Now run command “clear” and find the results of `num1` to `num4` again from the **workspace window**. Briefly explain what happened.
10. Solve questions 2, 4, 6, 8, 10, and 16 of chapter 1 from the textbook.

Lab Demonstration Requirements (40%):

Demonstrate at least 4 questions of step 10 above in the lab computer during the lab time.

After Lab Practice Questions (20%):

1. **Tracing Exercises:** Manually evaluate what will be the values of **x**, **y**, **j**, and **z** after the following MATLAB formatted instructions are executed (**Do these by hand: Do not use MATLAB in this practice!**). Must show all your calculation procedures.

```
aa = 130 + 4;  
bb = 5 / 10 + 2 - 4;  
cc = 5 % 2 * 11;  
x = aa - (bb * cc);  
m1 = 2 ^ 4 - 10;  
k1 = 3 ^ (5 - 11);  
f1 = 87 - 2 / 20;  
y = m1 * k1 - 10 - f1 / 3 ;  
dd = 4.5 * 2.3 - 10.3;  
ee = 5.3 + 6.7 - 0.9;  
j = abs (ee) + round (ee)*9 - fix (ee)*10;  
z = ceil (ee) - sqrt (dd*dd) * floor (ee) - j;
```

Lab Report Submission Requirement (40%):

Show the demonstrations as required on the lab computers and submit Lab #1 based on the provided templates on the Blackboard. Please post all the screenshots of the inputs and running results of each step on the lab manual.